

# A Market World Model for Public LLMs: Complete, Honest, Auditable Crypto Cognition before Decision

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## Abstract

Most quant stacks start from *strategy*. Public LLMs (GPT-, DeepSeek-, GLM-, and Gemini-class agents) that act in cryptocurrency markets fail earlier: they lack a complete, honest, auditable *market cognition base*. We present an anonymized *financial world-model runtime* that compiles asynchronous multi-band evidence into a point-in-time (PIT) world bundle. Epistemic observations  $O_{j,t} = (x, \tau, q, g, r)$  and compilation  $\Pi_t$  yield  $\mathcal{F}_t^{\text{AI}}$ ; completeness, honesty ( $B, U, H$ ), and WMI/ACWMI support a typed abstention contract; evidence IDs make actions auditable. Four live public LLMs (temperature 0) on the same 100 OOS asset-days per arm show a three-arm behavior ladder: Compiled thin-world abstain 1.0; Ungated (same content, no hard flag) mean 0.68 (range 0.43–0.86); Raw 0.04–0.75. Disclosure helps but is vendor-dependent; the typed boolean enforces full cross-vendor refusal, and models that act under Ungated lose money on thin support. A non-trading grounding workflow shows the same asymmetry for analysis: with the compiled three-band bundle, models recover ready/missing sets and tilt signs near-perfectly (mean F1/acc. 0.97); from a raw feed the same scores collapse to 0.04–0.23. A within-archive dense-world split shows the content probe is strongest on fully ready days. Economic CE remains a secondary nonempty-content probe—not a strategy Holy Grail. We detail the runtime architecture (collectors, vintage store, BandPIT clock, bundle schema, LLM adapter) as a systems contribution for ICAIF.

## CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics*.

## Keywords

world models, public LLMs, AI agents, cryptocurrency, point-in-time, abstention, trustworthy AI, market cognition, systems

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## 1 Introduction

Public LLMs are increasingly asked to analyze or trade crypto markets. The bottleneck is not another alpha signal. It is *understanding*: before decision, an agent needs a market world that is **complete** (missingness disclosed), **honest** (stale or role-incoherent evidence gated), and **auditable** (actions bound to evidence). Venue fragmentation, derivatives, on-chain flows, news, and macro vintages arrive on incompatible clocks [11, 13]. Generative world models [5, 6, 9]

and tool-using finance agents [18–20] typically assume a usable observation stream. Crypto agents often do not have one.

*Thesis.* We build a market *world-model runtime* for public LLMs: compile raw multi-band evidence into a PIT-safe cognition bundle that GPT / DeepSeek / GLM / Gemini-class models can call. Strategy metrics are secondary probes that the world has content—not the product objective. In slogan form: *quant from strategy; this runtime from understanding*.

*What we claim (and do not).* RQ1 tests whether a typed world interface can *enforce* world-conditional abstention across live public LLMs—a runtime contract, not a claim that models spontaneously discover thinness from raw ticks. RQ2 tests whether compiled band fields are economically nonempty versus same-input momentum. We do not claim unconditional trading alpha or a Dreamer-style generative simulator.

## Contributions.

- (1) **Cognition semantics.** Epistemic observations, compilation  $\Pi_t$ , a lag reconstruction bound, completeness/honesty fields, WMI/ACWMI abstention, evidence-bound actions.
- (2) **Anonymized runtime architecture.** Vintage-aware collectors, BandPIT previous-close clock, quality-tagged bundles, availability shocks  $O_t$ , OpenAI-compatible multi-vendor adapter, and a frozen Compiled / Ungated / Raw consumer protocol.
- (3) **Live three-arm validation.** Four live public LLMs on identical 100-day samples: Compiled enforces thin-world abstain 1.0; Ungated soft judgment averages 0.68; Raw often trades. A grounding workflow shows compiled worlds make ready/missing/tilt answers verifiable (mean F1/acc. 0.97 vs. 0.04–0.23 raw); a secondary content probe—including a dense-world split—shows vintaged band fields are economically nonempty.

## 2 Related Work

*World models in AI.* World models learn compact states and often dynamics [5, 6, 9]. We address the complementary *observation-side* problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy. We claim the missing systems layer generative WMs usually assume—not a Dreamer/JEPA simulator.

*LLM agents in finance.* ChatGPT return prediction [12], open financial LLMs [17], memory-augmented trading agents [10, 19], and tool-augmented multimodal agents [20] advance agent skill. Agentic tool use [18] still assumes aligned inputs. We evaluate within-model Compiled vs. Ungated vs. Raw under a frozen schema so the estimand is the *world interface*, not a new memory module or prompt trick.

*Selective prediction and trustworthy AI.* Abstention can be optimal under noise [2, 3]. We tie refusal to world quality via an explicit runtime contract, not classifier confidence alone.

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*PIT measurement and crypto microstructure.* Macro vintages [1] and crypto microstructure [11, 13] motivate multi-band compilation. Bootstrap discipline for secondary probes follows [7, 15, 16]. Asset-pricing ML [4, 8, 14] is adjacent: we validate nonempty compiled content for agents, not priced factors.

### 3 Market World-Model Semantics

DEFINITION 1 (EPISTEMIC OBSERVATION). *An epistemic observation is*

$$O_{j,t} = (x_{j,t}, \tau_{j,t}, q_{j,t}, g_{j,t}, r_{j,t}) :$$

*value, latest-available time, quality, main-view gate, semantic role. A bare feature dump that omits  $(\tau, q, g, r)$  is not a world observation.*

DEFINITION 2 (FINANCIAL WORLD STATE).  $W_t = \Pi_t(\mathcal{F}_t^{\text{raw}})$  with scalar quality  $q(W_t)$  (e.g. WMI/ACWMI) gating abstention;  $\mathcal{F}_t^{\text{AI}} = \sigma(W_t)$ .

DEFINITION 3 (COMPILATION).  $\Pi_t = B_t \circ M_t \circ A_t$ : align clocks, apply honesty/missingness gates, assemble band views.

*Completeness, honesty, quality.* Ready/limited/missing counts disclose coverage.  $\text{WMI}_t = B_t U_t H_t$ ; ACWMI supports regime-conditional gating. The Compiled contract exposes the abstain flag (should\_ai\_abstain) and thin\_world as first-class fields.

ASSUMPTION 1 (BOUNDED LAG AND INCREMENTS). *Band  $j$  has max observation lag  $\Delta_j$ ; latent state increments satisfy  $\|S_u - S_{u-1}\| \leq \bar{\delta}$ .*

PROPOSITION 1 (LAG RECONSTRUCTION BOUND). *Under Assumption 1,*

$$\|\tilde{S}_t - S_t\| \leq C_{\text{del}}\bar{\delta} + C_{\text{noi}}\varepsilon_t + C_{\text{miss}}m_t.$$

*Compilation cannot erase delay; it can refuse to pretend the bound is zero.*

*Intuition.* Crypto agents that concatenate tool dumps silently absorb delay and missingness into the feature vector. Prop. 1 makes those terms explicit: a thicker raw payload can worsen the reconstruction if honesty collapses.

PROPOSITION 2 (COMPILATION  $\neq$  FEATURE DUMP). *Enlarging raw span without  $\Pi_t$  need not enlarge usable  $\mathcal{F}^{\text{AI}}$ : ungated stale evidence can raise volume while lowering honesty/stability.*

PROPOSITION 3 (WORLD-CONDITIONAL ABSTENTION). *If non-abstain actions exceed a world-dependent cost  $c_{\text{abs}}(W_t)$ , the Bayes action is abstain; with costs decreasing in WMI, abstention forms a lower set in world quality [2].*

PROPOSITION 4 (LOBO CONTENT VS. GATING). *Deleting band  $j$  changes value through a content channel and a gating channel. Empirically we report content share under an ungated rule so the probe speaks to nonempty fields.*

*Reading for systems reviewers.* Props. 2–3 justify why the runtime exports gates and abstain flags rather than only tilts. Prop. 4 justifies the secondary economic probe: if deleting a durable band does not change actions, the bundle is decorative.

REMARK 1 (NOT A GENERATIVE WM). *We do not learn  $p(s_{t+1} | s_t, a_t)$ . We deliver a state compiler + abstention runtime.*

**Table 1: Anonymized module map (systems view).**

| Module            | Responsibility                                        |
|-------------------|-------------------------------------------------------|
| Collectors        | Ingest multi-band series with timestamps / vintages   |
| Vintage stores    | Persist raw, market, and analytics objects            |
| BandPIT compiler  | Previous-close readiness, $\Pi_t$ , WMI/ACWMI         |
| Bundle builder    | Assemble complete/honest/auditable JSON world state   |
| Shock index $O_t$ | Query availability outages for quasi-exogenous levers |
| LLM adapter       | OpenAI-compatible chat; frozen prompts; temp. 0       |
| Eval harness      | Compiled/Ungated/Raw sampling, transcripts, tables    |

## 4 Runtime Architecture

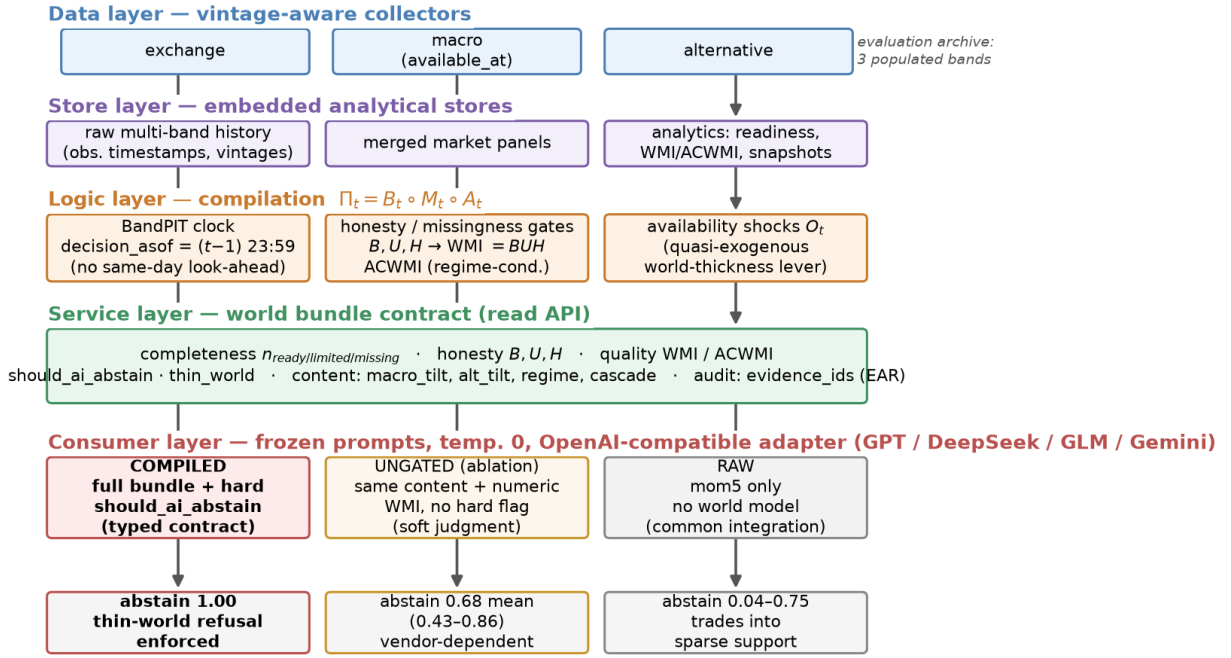
We describe the anonymized prototype as a layered systems object (Figure 1). Source paths and product names are omitted for double-blind review; an anonymized artifact will be released upon acceptance.

### 4.1 Layered design

- (1) **Data layer (collectors).** Exchange, macro, and alternative collectors write vintage-aware history: observation timestamps plus macro available\_at vintages.
- (2) **Store layer.** Embedded analytical stores hold raw series, merged market panels, and pipeline outputs (readiness, WMI/ACWMI, paper world snapshots).
- (3) **Logic layer (compilation).** BandPIT reconstructs band readiness on a previous-close clock; honesty/missingness gates and band assembly implement  $\Pi_t$ ; world-quality indices and availability shocks  $O_t$  are first-class.
- (4) **Service layer.** A read API serves quality-tagged AI bundles and health/readiness objects to consumers without requiring collector restarts for newly arrived rows.
- (5) **Consumer layer.** Frozen-prompt LLM / rule agents call an OpenAI-compatible adapter (temperature 0) under Compiled, Ungated, or Raw treatments.

### 4.2 Data flow: inputs, processing, outputs

Table 2 traces the runtime end-to-end. *Inputs:* the evaluation archive populates three public bands—exchange klines/funding/open-interest via REST, daily macro risk indices (equity volatility, dollar index) with available\_at vintages, and aggregate stablecoin circulation—each row stored with its observation timestamp. *Processing:* for every asset-day, BandPIT selects the latest observation with timestamp  $\leq (t-1)$  23:59, grades it fresh/stale/missing against per-band age thresholds, aggregates breadth/stability/honesty into WMI and regime-conditional ACWMI, derives content tilts (5-day macro-index changes; 7-day stablecoin-flow slope; return momentum), and logs availability shocks  $O_t$ . *Outputs:* a ~3,990-row PIT panel (10 assets  $\times$  399 days), per-day world bundles scoped to these three archive bands (Listing 1), and the consumer transcripts/tables behind Sections 5.



**Figure 1: Anonymized runtime architecture. Five layers map raw multi-band evidence to a typed world-bundle contract; the consumer layer runs the frozen three-arm protocol (Compiled / Ungated / Raw), whose live abstention outcomes are shown at the bottom.**

**Table 2: Data flow through the runtime (three-band evaluation archive).**

| Band        | Fetches (raw)                                           | Emitted (bundle)                                                            |
|-------------|---------------------------------------------------------|-----------------------------------------------------------------------------|
| exchange    | OHLCV klines (1h/1d), funding, open interest, orderbook | mom5; band status/age; funding context                                      |
| macro       | equity-vol and dollar indices; vintaged macro series    | macro_tilt (5d changes); status/age                                         |
| alternative | aggregate stablecoin circulation                        | alt_tilt (7d flow slope); status/age                                        |
| payoff      | daily closes (external reference)                       | close-to-close $r_t$ for scoring only                                       |
| derived     | —                                                       | $B, U, H$ , WMI, ACWMI, should_ai_abstain, thin_world, $O_t$ , evidence_ids |

### 4.3 PIT previous-close clock

For calendar day  $t$ , the decision information set is reconstructed at  $(t-1)$  23:59. The payoff is the same-calendar-day close-to-close return  $r_t$ . This removes same-day look-ahead in band statuses and vintaged macro/alternative content. The bundle records the clock in decision\_asof, and every live run pins timing\_protocol to the previous-close protocol.

**Table 3: World bundle field groups (cognition base).**

| Group        | Fields                                      |
|--------------|---------------------------------------------|
| Timing       | decision_asof, previous-close protocol      |
| Completeness | $n_{ready/limited/missing}$ , band statuses |
| Honesty      | $B, U, H$ , main-view / stale gates         |
| Quality      | WMI, ACWMI, should_ai_abstain, thin_world   |
| Content      | macro_tilt, alt_tilt, regime, cascade       |
| Audit        | evidence_ids, EAR required                  |

### 4.4 World bundle contract

Table 3 lists field groups. Listing 1 shows a compact Compiled example from the OOS panel: exchange missing so 2/3 archive bands ready, WMI low, should\_ai\_abstain=true, and evidence\_ids binding actions to disclosed bands/tilts. Ungated removes the hard boolean (and threshold) but keeps numeric WMI/ACWMI and completeness. Raw retains only mom5 (plus a non-informative noise bit).

**Listing 1: Compact Compiled bundle example (OOS exchange gap).**

```
{
  "date": "2026-05-11",
  "asset": "ADA",
  "decision_asof": "2026-05-10T23:59:00",
  "completeness": {
    "n_ready": 2,
    "n_missing": 1,
    "ready_share": 0.67,
    "archive_bands": [
      "exchange", "macro", "alternative"
    ]
  },
  "world_model_index": {
    "wmi": 0.015,
    "acwmi": 0.30,
    "should_ai_abstain": true,
    "thin_world": true
  },
  "macro_tilt": 1.0,
  "alt_tilt": -1.0,
}
```

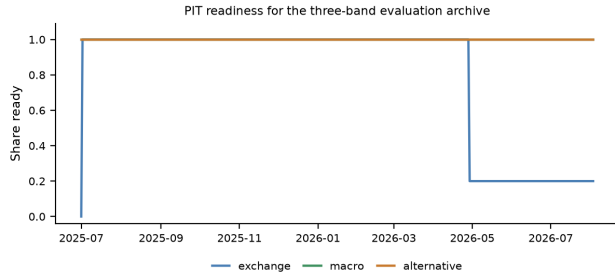


Figure 2: PIT readiness for the three-band evaluation archive.

```
"detected_regime": "crisis",
"audit": {
  "evidence_ids": [
    "band:exchange:missing",
    "band:macro:ready",
    "band:alternative:ready",
    "tilt:macro:1.0"
  ]
}
```

#### 4.5 Three information arms and LLM adapter

- **Compiled (product contract):** full bundle; frozen prompt *requires* abstain when should\_ai\_abstain is true.
- **Ungated (ablation):** same content/completeness/numeric WMI; no hard flag; soft prompt asks the model to judge thinness.
- **Raw (without world model):** mom5 only; no quality index or abstention guidance.

Actions  $\in \{\text{bullish, bearish, neutral, abstain}\}$ . Understanding metrics prioritize thin-world abstain rate and EAR proxy over CE. The evaluation archive is the three-band set {exchange, macro, alternative} (Figure 2): macro and alternative stay ready throughout; exchange readiness drops in the later OOS window and defines the within-archive thickness contrast below. The adapter speaks OpenAI-compatible chat completions; model IDs are passed through; credentials stay in environment variables (never in artifacts).

#### 4.6 Frozen prompt contracts

Listings 2–3 excerpt the frozen Compiled and Ungated contracts (temperature 0; action schema identical). The only intentional difference is hard vs. soft abstention. Raw prompts omit world-quality language entirely.

Listing 2: Compiled contract (excerpt).

```
Decision rules (excerpt):
1. If world_model_index.should_ai_abstain is true
   OR the world is thin, you MUST "abstain".
2. Prefer band content over raw momentum
   (macro_tilt, alt_tilt are PIT-safe).
3. Rationale must be consistent with evidence_ids.
```

Listing 3: Ungated soft contract (excerpt).

```
Decision guidance (soft):
1. Judge whether missingness / low WMI/ACWMI
   makes action unsafe; you MAY "abstain".
2. Prefer band content when you act.
NO hard abstain boolean; NO forced policy.
```

Not “just instruction following.” Compiled abstention is partly prompt obedience—and that is the *product requirement*: production agents need an enforceable refuse interface when  $q(W_t)$  is low (Prop. 3). Ungated measures soft judgment from disclosure alone; Raw measures the common “price signal only” integration. Reliable

Table 4: Live three-arm ladder on identical 100 OOS asset-days (Wilson 95% CIs).  $\Delta$ CE secondary.

| Model                 | Thin C | Thin U [CI]    | Abs. R [CI]    | $\Delta$ CE |
|-----------------------|--------|----------------|----------------|-------------|
| gpt-5.4-mini          | 1.00   | 0.63 [.53,.72] | 0.04 [.02,.10] | +0.73       |
| deepseek-v4-flash     | 1.00   | 0.81 [.72,.87] | 0.75 [.66,.82] | +1.22       |
| glm-5.2               | 1.00   | 0.86 [.78,.91] | 0.11 [.06,.19] | +1.04       |
| gemini-3.5-flash-lite | 1.00   | 0.43 [.34,.53] | 0.07 [.03,.14] | +1.30       |
| Mean                  | 1.00   | 0.68           | —              | +1.07       |

Compiled refusal is the SLA; spontaneous Ungated refusal is a bonus.

#### 4.7 Service surface and reproducibility

The service layer exposes read-only objects used by the consumer harness: (i) quality-tagged world bundles; (ii) band readiness / WMI–ACWMI snapshots; (iii) availability-shock queries  $O_t$ ; (iv) health metadata disclosing whether paper engines or production proxies hydrate ACWMI inputs. Collectors and the logic pipeline can run as supervised jobs; the API reads stores live, so newly compiled rows appear without process bounce. For review we freeze prompts, temperature, action schema, IS/OOS cut, and sampling seed; model IDs and an OpenAI-compatible base URL are configuration, not estimands. Credentials never enter committed artifacts. Upon acceptance we will release an anonymized repository with the consumer harness, frozen prompts, and table-reproduction scripts.

### 5 Experiments

#### 5.1 Setup

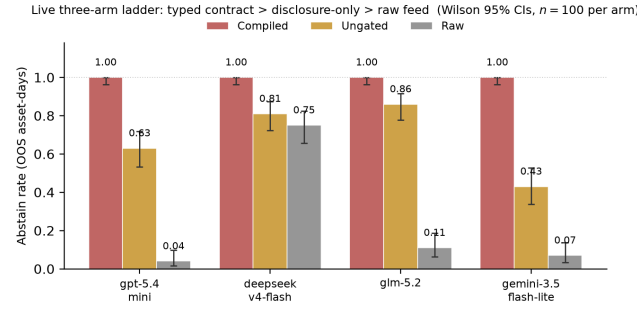
PIT panel  $\sim 399$  days ( $\approx 13$  months), 10 liquid names, previous-close clock, chronological IS/OOS 200/200 days. Live RQ1: all three arms run on the *same* stratified 100 OOS asset-days (same sampling seed), temperature 0, frozen prompts, OpenAI-compatible gateway. Models: gpt-5.4-mini, deepseek-v4-flash, glm-5.2, gemini-3.5-flash-lite. Offline mocks reproduce the protocol on full OOS without keys. Scarce/thin labels bind on OOS by design: primary test is honesty under sparse support, not a dense-world trading horse-race.

#### 5.2 RQ1 (primary): Three-arm live behavior

Table 4 and Figure 3 are the headline result; all arms share the same 100 OOS asset-days. Under Compiled, all four models abstain on every thin-world day (thin-abs. 1.0); rationales cite world fields on 93–99% of days. Under Ungated, mean thin-abs. falls to 0.68 (range 0.43–0.86): DeepSeek and GLM stay above 0.80, GPT drops to 0.63, and gemini-3.5-flash-lite falls to 0.43 and trades. When Ungated models do act, they lose (CE  $-1.04$  to  $+0.02$  across models)—soft judgment fails exactly where enforcement matters. Under Raw, abstain rates are 0.04–0.75 with substantial directional mass (Table 5). Mean  $\Delta$ CE (Compiled–Raw) =  $+1.07$  is reported last: Compiled CE is zero under full abstention, so the gap mainly reflects Raw overtrading—not a Holy Grail.

*Inside the Ungated arm.* Ungated rationales cite disclosed world fields (missingness, readiness, WMI) on 53–93% of days, so the





**Figure 3: Live three-arm ladder (same 100 days per arm; Wilson 95% CIs). Typed contract > disclosure-only > raw feed; soft judgment is vendor-dependent.**

**Table 5: Raw-arm action mix (100 days). Compiled is 100% abstain.**

| Model                 | Abs. | Bull. | Bear. | Neut. |
|-----------------------|------|-------|-------|-------|
| gpt-5.4-mini          | 0.04 | 0.28  | 0.38  | 0.30  |
| deepseek-v4-flash     | 0.75 | 0.14  | 0.04  | 0.07  |
| glm-5.2               | 0.11 | 0.40  | 0.48  | 0.01  |
| gemini-3.5-flash-lite | 0.07 | 0.44  | 0.49  | 0.00  |

models demonstrably *read* the world; they differ in acting on it. Non-abstain Ungated actions are overwhelmingly directional (e.g. Gemini: 55 bearish, 2 bullish of 57 trades) and destroy value on thin support—evidence that vendor-specific caution, not lack of disclosure, drives the gap the typed contract closes.

*Offline diagnostics.* World-honoring offline followers abstain at 1.0 on full OOS; momentum-only and noisy mocks ignore thin-world flags (thin-abs. 0)—the Raw failure mode in miniature. Offline results are protocol checks; live vendors are the product claim.

### 5.3 RQ3: Non-trading cognition—grounding workflow

The cognition base should serve analysis, not only trade/abstain decisions. We run a live multi-step *grounding workflow* on 50 OOS asset-days per arm. Given Compiled or Raw inputs, the model must (i) list which of {exchange, macro, alternative} are ready, (ii) list which are missing, (iii) report  $\text{sign}(\text{macro\_tilt})$  and  $\text{sign}(\text{alt\_tilt})$ , and (iv) judge sufficiency. Steps (i)–(iii) are scored against PIT ground truth. Table 6: with the Compiled bundle, mean ready-band F1 0.97, missing-band F1 0.97, and tilt-sign accuracy 0.97; with Raw, the same scores collapse to 0.04, 0.23, and 0.19. Sufficiency alone is *not* the discriminator once completeness is scoped to real archive bands—models may call a 3/3-ready day “sufficient” while WMI still marks the world thin—so the product claim is *verifiable epistemics*: only the compiled world lets the model say what is ready, what is missing, and which tilts are present, and be checked. This is the analyst-facing counterpart of Prop. 1’s missingness term.

**Table 6: Grounding workflow (50 days): ready / missing / tilt signs.**

| Model                 | Ready F1 |      | Missing F1 |      | Tilt-sign acc. |      |
|-----------------------|----------|------|------------|------|----------------|------|
|                       | C        | R    | C          | R    | C              | R    |
| gpt-5.4-mini          | 0.94     | 0.00 | 0.94       | 0.31 | 0.94           | 0.20 |
| deepseek-v4-flash     | 0.98     | 0.06 | 0.98       | 0.20 | 0.98           | 0.18 |
| glm-5.2               | 0.98     | 0.09 | 1.00       | 0.11 | 0.98           | 0.20 |
| gemini-3.5-flash-lite | 0.96     | 0.03 | 0.96       | 0.30 | 0.96           | 0.18 |
| Mean                  | 0.97     | 0.04 | 0.97       | 0.23 | 0.97           | 0.19 |

**Table 7: Secondary content probe (not the LLM product claim).**

| Policy / contrast       | Sharpe | CE                  |
|-------------------------|--------|---------------------|
| Momentum always         | 0.101  | −0.202              |
| Thick ungated (content) | 0.767  | 0.132               |
| Mechanism – Momentum    | —      | 0.334 ( $p=0.034$ ) |

**Table 8: Dense-world split of the content probe (OOS; three archive bands).**

| Slice               | Share | Mech CE | $\Delta\text{CE}$ vs mom |
|---------------------|-------|---------|--------------------------|
| All OOS             | 1.00  | 0.130   | 0.336                    |
| Thick (3/3 ready)   | 0.61  | 0.059   | 0.316                    |
| Thin (exchange gap) | 0.39  | 1.048   | 0.129                    |

### 5.4 RQ2: Nonempty compiled content (incl. dense-world split)

A transparent rule shares return inputs with momentum and adds vintaged  $\text{macro\_tilt}$  /  $\text{alt\_tilt}$ . Pre-specified OOS contrast mechanism – momentum:  $\Delta\text{CE}=0.334$  ( $p=0.034$ ; stationary  $p=0.022$ ). LOBO (Prop. 4): deleting macro or alternative collapses to momentum (content share 1.0 under the ungated rule). Relative gaps survive 10–25bps costs; absolute CE is cost-fragile. A 2017–2026 audit without vintaged archives finds no hidden return-rule alpha.

*Within-archive dense world.* Among the three archive bands, 61% of OOS asset-days are *thick* (exchange+macro+alternative all ready) and 39% are *thin* (exchange gap; Figure 2). Table 8: the same mechanism–momentum gap remains economically meaningful on thick days ( $\Delta\text{CE}=0.316$ ) and shrinks when exchange support is absent ( $\Delta\text{CE}=0.129$ ). Dense support does not invent a Holy Grail; it shows the compiled content channel is strongest precisely when the archive is fully populated.

### 5.5 Discussion and threats

The live ladder  $\text{Abs}_C = 1.0 > \text{Abs}_U \approx 0.68 > \text{Abs}_R \in [0.04, 0.75]$  is the product evidence: a cognition base that marks the world thin induces vendor-stable refusal; disclosure alone is incomplete; raw feeds over-act. Cross-vendor heterogeneity under Ungated (GLM 0.86 vs. Gemini 0.43, non-overlapping Wilson CIs) is exactly why a

typed contract is valuable in production: soft judgment is model-dependent, while Compiled refusal is stable. RQ2 shows band fields are not decorative. Claim: *understanding first, strategy second*.

Relative to FinMem / FinAgent-style systems [19, 20], we do not propose a new memory or tool router. We propose the observation compiler those agents implicitly need when the market feed is asynchronous and quality-heterogeneous. Relative to return-prediction LLM studies [12], we score refusal and auditability before PnL.

*Threats to validity.* (1) *Prompt contract*—Compiled refusal is partly instructed; Ungated is the soft contrast. (2) *Three-band archive*—evaluation uses exchange/macro/alternative only; within-archive thickness is not a multi-domain dense world. (3) *Baselines*—Raw matches common thin integrations; Ungated is the same-content control. (4) *Sample size*—100 live days per arm (Wilson CIs reported); harness supports full ~2000-row OOS sweeps. (5) *Vendor routing*—gateway and model IDs vary; prompts/schema are frozen for within-model contrasts. (6) *Task scope*—the trading action space is a coarse directional schema; the grounding workflow (Sec. 5.3) covers ready/missing/tilts, not full analyst research.

## 6 Design Rationale for ICAIF Systems Readers

Three implementation choices are deliberate.

*Export gates, not only features.* Silent feature stores hide staleness. By placing completeness, honesty, and `should_ai_abstain` in the bundle, the runtime makes refusal a first-class machine-readable event rather than a free-text apology.

*Previous-close PIT over wall-clock streaming.* Intraday streaming demos look impressive but confound look-ahead. The previous-close clock is conservative and auditable: every live decision cites a `decision_asof` stamp that a reviewer can recompute from the panel.

*Three arms instead of one leaderboard.* A single Compiled-vs-holdout Sharpe invites Holy-Grail readings. Compiled / Ungated / Raw separates (i) typed enforcement, (ii) soft judgment from disclosure, and (iii) no world model—matching how teams actually integrate LLMs.

## 7 Limitations

We do not claim generative dynamics or unconditional trading alpha. The evaluation archive covers three public bands (exchange, macro, alternative); richer domains (news, on-chain, options) are out of scope for this study and are not claimed as censored feeds. Dense-world selective-prediction horse-races beyond within-archive thickness remain future work. Live samples are budget-constrained (100 asset-days per arm); the harness already supports full OOS replication. Camera-ready will restore author identity and point to an anonymized repository deferred during review.

## 8 Conclusion

Public LLMs need a market to *understand* before they decide. We contribute cognition semantics and an anonymized layered runtime—collectors, vintage store, BandPIT compilation, bundle contract, and LLM adapter—that turns asynchronous crypto evidence into a complete, honest, auditable world state. Live public LLMs exhibit a

three-arm ladder: Compiled enforces thin-world refusal, Ungated partially recovers it from disclosure alone, and Raw often trades into sparse support. The same bundle makes analyst-facing answers verifiable (grounding ready/missing/tilt mean 0.97 vs. 0.04–0.23 raw), and content probes—including a dense-world split—confirm the compiled state is nonempty. Strategy remains downstream of understanding.

## References

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